

From Jacobian to Hessian

Stationarity, Off-Shell Residuals and the Deformation Theory of Einstein Metrics*

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Abstract

The Hessian of an action is the Jacobian of its Euler–Lagrange map. This identity opens a clean route from elementary multivariable calculus to linearised general relativity. The first variation selects a stationary background; the Jacobian of that stationary map or equivalently the Hessian of the action, determines infinitesimal deformations, while its kernel, image and cokernel organise physical modes, correctable residuals and obstructions in turn. Higher variations test whether infinitesimal deformations continue to nearby exact solutions. Off-shell residuals fit naturally into this picture as sources for a Newton-type correction; they do not, by themselves, identify new matter. The final sections examine why moving a modified gravitational operator to the source side is only a change of representation, and why a tempting equations-of-motion-squared modification is perturbatively redundant by field redefinition and, treated non-perturbatively, generically introduces additional propagating modes.

1 A stationary point which is not a minimum

Begin with the surface

$$f(x, y) = \frac{1}{2}x^2 - \frac{1}{2}y^2. \quad (1)$$

It rises in the x -direction and falls in the y -direction. Its gradient is

$$\nabla f(x, y) = \begin{pmatrix} x \\ -y \end{pmatrix}. \quad (2)$$

At the origin,

$$\nabla f(0, 0) = 0. \quad (3)$$

Thus the origin is stationary, although it is neither a minimum nor a maximum: it is a saddle point. This is the first distinction needed later in gravity. Stationarity means that the first-order change vanishes. It does not mean that the action has been minimised.

Now regard the gradient itself as a map,

$$\nabla f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \quad (x, y) \longmapsto (x, -y). \quad (4)$$

If the point is displaced by

$$h = \begin{pmatrix} h_x \\ h_y \end{pmatrix}, \quad (5)$$

*For a complementary treatment in terms of contact jet bundles, prolongation and integrability conditions—especially in metric-affine gravity—see McCulloch-James [1].

then

$$\nabla f((x, y) + h) - \nabla f(x, y) = \begin{pmatrix} h_x \\ -h_y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} h. \quad (6)$$

The matrix

$$J_{\nabla f} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (7)$$

is the Jacobian of the gradient map. Because the map being differentiated is itself the first derivative of a scalar function, this Jacobian is also the Hessian of f :

$$H_f = J_{\nabla f} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (8)$$

For this quadratic example the Taylor expansion is exact:

$$f((x, y) + h) = f(x, y) + \nabla f(x, y) \cdot h + \frac{1}{2} h^\top H_f h. \quad (9)$$

At the stationary point,

$$f(h_x, h_y) - f(0, 0) = \frac{1}{2} h_x^2 - \frac{1}{2} h_y^2. \quad (10)$$

The Hessian therefore records the local stationary structure: one direction curves upward and the other curves downward.

The gradient is the stationary-equation map. Its Jacobian tells us how that equation changes when the underlying point is displaced. That Jacobian is the Hessian of the original scalar function.

2 From the saddle to the general Jacobian

The saddle function, $f_0(x, y) = \frac{1}{2}x^2 - \frac{1}{2}y^2$, has a gradient,

$$\nabla f_0(x, y) = \begin{pmatrix} x \\ -y \end{pmatrix}, \quad (11)$$

which is a particular example of a differentiable map between vector spaces.

For a general differentiable map

$$F : \mathbb{R}^n \longrightarrow \mathbb{R}^m, \quad (12)$$

with components $F^a(x)$, the Jacobian is

$$(J_F)^a{}_i(x) = \frac{\partial F^a}{\partial x^i}. \quad (13)$$

It gives the first-order response of the map:

$$F(x + h) = F(x) + J_F(x)h + O(\|h\|^2). \quad (14)$$

Thus the Jacobian is the tangent map of F . It converts an infinitesimal displacement in the domain into the corresponding first-order displacement in the codomain:

$$h \longmapsto J_F(x)h. \quad (15)$$

When the map is a gradient

Now let

$$f : \mathbb{R}^n \longrightarrow \mathbb{R} \quad (16)$$

be an arbitrary twice-differentiable scalar function. Its gradient is itself a map,

$$\nabla f : \mathbb{R}^n \longrightarrow \mathbb{R}^n. \quad (17)$$

The Jacobian of this particular map is the Hessian of f :

$$H_f(x) = J_{\nabla f}(x) = \left(\frac{\partial^2 f}{\partial x^i \partial x^j} \right). \quad (18)$$

Hence the Hessian is simultaneously the second derivative of the scalar function f ; and first derivative of its gradient map ∇f .

For the original saddle function f_0 , this gives

$$H_{f_0} = J_{\nabla f_0} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (19)$$

For an arbitrary f , the second-order Taylor expansion is

$$f(x+h) = f(x) + \nabla f(x) \cdot h + \frac{1}{2} h^\top H_f(x) h + O(\|h\|^3). \quad (20)$$

Meanwhile, the first-order expansion of the gradient map is

$$\nabla f(x+h) = \nabla f(x) + H_f(x)h + O(\|h\|^2). \quad (21)$$

These are two views of the same matrix. In the first equation, H_f determines the quadratic change in the scalar function. In the second, it determines the linear change in its gradient.

At a stationary point $\bar{x}$,

$$\nabla f(\bar{x}) = 0. \quad (22)$$

The vector $\nabla f(x)$ may therefore be regarded as the residual of the stationary equation, and

$$H_f(\bar{x})h \quad (23)$$

is its first-order response to the displacement h .

The Hessian is the Jacobian of the stationary-equation map $x \mapsto \nabla f(x)$. It converts a displacement of the configuration into the corresponding first-order change of the stationary residual.

At a stationary point, the signs of the eigenvalues of H_f describe the local quadratic shape. Positive eigenvalues identify minimum-like directions, negative eigenvalues identify maximum-like directions, and zero eigenvalues identify directions in which the quadratic approximation is flat.

Remark. The family

$$f_\lambda(x, y) = \frac{1}{2}x^2 + \frac{\lambda}{2}y^2 \quad (24)$$

places these possibilities in one elementary model:

$$H_{f_\lambda} = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix}. \quad (25)$$

For $\lambda > 0$, the origin is a minimum; for $\lambda < 0$, it is a saddle; and for $\lambda = 0$, the Hessian has a zero direction. The case $\lambda = -1$ recovers the original saddle f_0 . The case $\lambda = 0$ anticipates the degenerate directions that later arise from gauge symmetry.

The Jacobian of a general equation

The preceding construction extends beyond gradient maps. Suppose

$$F(x) = 0 \tag{26}$$

is a system of equations, and let $x(\varepsilon)$ be a curve of exact solutions satisfying

$$x(0) = \bar{x}, \quad h = \dot{x}(0). \tag{27}$$

Differentiating $F(x(\varepsilon)) = 0$ gives

$$0 = \left. \frac{d}{d\varepsilon} F(x(\varepsilon)) \right|_{\varepsilon=0} = J_F(\bar{x})h. \tag{28}$$

Consequently,

$$h \in \ker J_F(\bar{x}). \tag{29}$$

The kernel of the Jacobian therefore contains the infinitesimal directions tangent to the solution set. When $F = \nabla f$, this statement becomes

$$H_f(\bar{x})h = 0. \tag{30}$$

Thus the kernel of the Hessian contains infinitesimal directions along which the stationary point may be deformed without violating the stationary equation at first order.

3 From the gradient map to the Euler–Lagrange map

For an ordinary scalar function, the gradient assigns to each point its stationary residual. Field theory has the same structure, with the action in place of the scalar function. Let $S[\phi]$ be an action, where ϕ denotes a field configuration. For a variation η , its first variation is

$$\delta S_\phi[\eta] = \left. \frac{d}{d\varepsilon} S[\phi + \varepsilon\eta] \right|_{\varepsilon=0}. \tag{31}$$

After integrations by parts, this takes the form

$$\delta S_\phi[\eta] = \langle E[\phi], \eta \rangle + \text{boundary terms}. \tag{32}$$

As ϕ varies, $E[\phi]$ defines the Euler–Lagrange map. It is the field-theoretic counterpart of the gradient map:

$$\boxed{\nabla f(x) \longleftrightarrow E[\phi]}. \tag{33}$$

Subject to the chosen boundary conditions, $\bar{\phi}$ is stationary precisely when

$$E[\bar{\phi}] = 0. \tag{34}$$

The brackets denote the natural integrated pairing between a field-equation residual and a field variation. For symmetric rank-two tensors on a metric background $\bar{g}$, we use the L^2 -type pairing

$$\langle A, B \rangle_{\bar{g}} = \int_M A^{\mu\nu} B_{\mu\nu} dV_{\bar{g}}, \quad dV_{\bar{g}} = \sqrt{|\bar{g}|} d^d x. \tag{35}$$

In Lorentzian signature this is an integrated bilinear pairing rather than a positive-definite inner product. Compact support, suitable fall-off conditions or appropriate boundary conditions are assumed whenever integrations by parts are used.

The functional Hessian

The derivative of the Euler–Lagrange map at ϕ is

$$DE_\phi[\eta] = \left. \frac{d}{d\varepsilon} E[\phi + \varepsilon\eta] \right|_{\varepsilon=0}. \quad (36)$$

It gives the first-order change of the field-equation residual:

$$E[\phi + \varepsilon\eta] = E[\phi] + \varepsilon DE_\phi[\eta] + O(\varepsilon^2). \quad (37)$$

Since the Euler–Lagrange map is the first derivative of the action, its derivative is the functional Hessian. For two admissible variations η and ζ ,

$$\delta^2 S_\phi[\eta, \zeta] = \langle \zeta, DE_\phi[\eta] \rangle_\phi. \quad (38)$$

In compact notation,

$$\boxed{\mathcal{H}_\phi = D^2 S_\phi = DE_\phi.} \quad (39)$$

After integrations by parts, and under the stated boundary conditions, the functional Hessian is formally self-adjoint:

$$\langle \zeta, \mathcal{H}_\phi \eta \rangle_\phi = \langle \eta, \mathcal{H}_\phi \zeta \rangle_\phi. \quad (40)$$

Formal self-adjointness does not imply that the quadratic form is positive or that the Hessian is invertible; in gravity it may be indefinite and possess a gauge kernel.

The Euler–Lagrange map is the field-theoretic stationary-equation map. Its Jacobian is the functional Hessian of the action.

An arbitrary differentiable field equation possesses a Jacobian. Its Jacobian is a Hessian in the variational sense only when the equation arises as the first variation of an action. The resulting Jacobi operators and deformation theory for critical metrics of geometric curvature functionals are developed systematically by Viaclovsky [2].

4 Exact solutions, on-shell backgrounds and linearisation

For general relativity, the dynamical field is the spacetime metric g . Define the Einstein equation map by

$$E_{\mu\nu}[g] = G_{\mu\nu}[g] + \Lambda g_{\mu\nu} - \kappa T_{\mu\nu}[g]. \quad (41)$$

For any proposed metric g , the tensor $E_{\mu\nu}[g]$ measures its failure to satisfy the full Einstein equation. An exact solution satisfies

$$E_{\mu\nu}[g] = 0. \quad (42)$$

Such a metric is said to be *on shell*. Thus

$$\boxed{g \text{ is on shell} \iff g \text{ satisfies the full nonlinear field equation.}} \quad (43)$$

A metric for which

$$E_{\mu\nu}[g] \neq 0 \quad (44)$$

is said to be *off shell*. It is still a perfectly legitimate metric at which the action and its derivatives may be evaluated, but it is not an exact solution of the field equation.

The word “shell” refers to the subset of the space of all possible fields selected by the field equation:

$$\mathcal{E} = \{g : E[g] = 0\}. \quad (45)$$

Metrics lying in $\mathcal{E}$ are on shell; metrics outside $\mathcal{E}$ are off shell.

Configuration	Meaning
$E[g] = 0$	g is an exact solution and is on shell
$E[g] \neq 0$	g has a nonzero field-equation residual and is off shell

Choosing a background

To study small departures from a known spacetime, choose a background metric $\bar{g}$ and write

$$g_{\mu\nu}(\varepsilon) = \bar{g}_{\mu\nu} + \varepsilon h_{\mu\nu} + O(\varepsilon^2). \quad (46)$$

Here $h_{\mu\nu}$ is the first-order metric perturbation. We first choose $\bar{g}$ to be an exact solution:

$$E_{\mu\nu}[\bar{g}] = 0. \quad (47)$$

The background is therefore on shell. This assumption is what allows the zeroth-order term in the perturbed equation to vanish. Expanding the Einstein map gives

$$E[\bar{g} + \varepsilon h] = E[\bar{g}] + \varepsilon DE_{\bar{g}}[h] + O(\varepsilon^2). \quad (48)$$

Because the background is on shell,

$$E[\bar{g}] = 0, \quad (49)$$

and hence

$$E[\bar{g} + \varepsilon h] = \varepsilon DE_{\bar{g}}[h] + O(\varepsilon^2). \quad (50)$$

If the perturbed metric is also to satisfy the Einstein equation to first order, the coefficient of ε must vanish:

$$\boxed{DE_{\bar{g}}[h] = 0}. \quad (51)$$

This is the linearised Einstein equation and it is important to distinguish the two statements

$$E[\bar{g}] = 0 \quad (52)$$

and

$$DE_{\bar{g}}[h] = 0. \quad (53)$$

The first says that the background metric is an exact solution of the nonlinear equation. The second says that h is an infinitesimal deformation which preserves that equation to first order.

The same expansion at the level of the action

The action expands about the background as

$$S[\bar{g} + \varepsilon h] = S[\bar{g}] + \varepsilon \delta S_{\bar{g}}[h] + \frac{\varepsilon^2}{2} \delta^2 S_{\bar{g}}[h, h] + O(\varepsilon^3). \quad (54)$$

Because $\bar{g}$ is an on-shell background, it is stationary:

$$\delta S_{\bar{g}}[h] = 0 \quad (55)$$

for every admissible h . The leading perturbative contribution is therefore the quadratic action

$$S_{\bar{g}}^{(2)}[h] = \frac{1}{2} \langle h, \mathcal{H}_{\bar{g}} h \rangle. \quad (56)$$

Varying this quadratic action with respect to h gives

$$\mathcal{H}_{\bar{g}} h = 0. \quad (57)$$

Since

$$\mathcal{H}_{\bar{g}} = DE_{\bar{g}}, \quad (58)$$

this is precisely

$$DE_{\bar{g}}[h] = 0. \quad (59)$$

Thus the same linearised equation has two complementary interpretations:

$DE_{\bar{g}}[h] = 0$ is the linearisation of the full field equation, $\mathcal{H}_{\bar{g}} h = 0$ is the stationary equation of the quadratic action.	(60)
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The background $\bar{g}$ is stationary with respect to the full action. The perturbation h is stationary with respect to the quadratic action obtained by expanding about $\bar{g}$.

Neither statement generally describes a minimum. In Lorentzian gravity the quadratic form is indefinite, and diffeomorphism symmetry produces gauge zero directions. Here “stationary” means that the relevant first variation vanishes, rather than that the action takes a least value.

5 Off-shell backgrounds: correction rather than deformation

Before treating an off-shell background, it is useful to distinguish two questions. The first is a deformation question:

Given an exact solution $\bar{g}$, which infinitesimal changes preserve the field equation to first order?

Since

$$E[\bar{g}] = 0, \quad (61)$$

the answer is

$$DE_{\bar{g}}[h] = 0. \quad (62)$$

The second is a correction question:

Given a configuration $\bar{g}$ which is not an exact solution, which change moves it towards the solution set?

This is the off-shell problem. Let

$$R = E[\bar{g}] \neq 0 \quad (63)$$

be the residual of the chosen background. Expanding the field equation at $\bar{g}$ gives

$$E[\bar{g} + h] = E[\bar{g}] + DE_{\bar{g}}[h] + \frac{1}{2} D^2 E_{\bar{g}}[h, h] + O(\|h\|^3). \quad (64)$$

Therefore

$$E[\bar{g} + h] = R + DE_{\bar{g}}[h] + O(\|h\|^2). \quad (65)$$

If $\bar{g} + h$ is to satisfy the field equation to the accuracy of the linear approximation, then

$$R + DE_{\bar{g}}[h] = 0. \quad (66)$$

Hence

$$\boxed{DE_{\bar{g}}[h] = -R.} \quad (67)$$

This is an inhomogeneous equation for the correction h . It differs from the homogeneous deformation equation

$$DE_{\bar{g}}[h] = 0 \quad (68)$$

obtained at an on-shell background.

Background	Linear equation	Interpretation
$E[\bar{g}] = 0$	$DE_{\bar{g}}[h] = 0$	Find infinitesimal directions tangent to the solution space.
$E[\bar{g}] = R \neq 0$	$DE_{\bar{g}}[h] = -R$	Find a first correction which cancels the background residual.

Why on shell is natural for deformation theory

Let

$$\mathcal{E} = \{g : E[g] = 0\} \quad (69)$$

be the space of exact solutions. If $\bar{g} \in \mathcal{E}$, then a curve of solutions $g(\varepsilon)$ satisfying

$$g(0) = \bar{g} \quad (70)$$

has tangent

$$h = \dot{g}(0). \quad (71)$$

Differentiating

$$E[g(\varepsilon)] = 0 \quad (72)$$

gives

$$DE_{\bar{g}}[h] = 0. \quad (73)$$

Thus

$$T_{\bar{g}}\mathcal{E} \subseteq \ker DE_{\bar{g}}, \quad (74)$$

with the usual qualifications concerning gauge directions, regularity and linearisation stability.

This construction requires $\bar{g} \in \mathcal{E}$. A point outside the solution space does not possess a tangent space to that solution space. At such a point, $DE_{\bar{g}}$ still exists, but it describes how the nonzero residual changes under a displacement.

From an equivariant viewpoint, the field equation defines a diffeomorphism-invariant solution set. Choosing an on-shell background means choosing a point of that invariant set and studying its local deformation structure. Choosing an off-shell background instead leads to a problem of approach or correction.

The action expanded off shell

The distinction can also be seen directly in the action. About an arbitrary background $\bar{g}$,

$$S[\bar{g} + h] = S[\bar{g}] + \delta S_{\bar{g}}[h] + \frac{1}{2} \delta^2 S_{\bar{g}}[h, h] + O(\|h\|^3). \quad (75)$$

If $\bar{g}$ is off shell, the linear term does not vanish:

$$\delta S_{\bar{g}}[h] = \langle R, h \rangle, \quad R = E[\bar{g}], \quad (76)$$

up to boundary terms. The action truncated at quadratic order is therefore

$$S_{\text{trunc}}[h] = S[\bar{g}] + \langle R, h \rangle + \frac{1}{2} \langle h, \mathcal{H}_{\bar{g}} h \rangle. \quad (77)$$

Requiring this truncated action to be stationary with respect to h gives

$$R + \mathcal{H}_{\bar{g}} h = 0. \quad (78)$$

Since

$$\mathcal{H}_{\bar{g}} = DE_{\bar{g}}, \quad (79)$$

we recover

$$DE_{\bar{g}}[h] = -R. \quad (80)$$

The linear term $\langle R, h \rangle$ is sometimes called a *tadpole*. Its presence records that the expansion has been made about a configuration which is not stationary for the full action.

Expanding on shell removes the linear term in the action and produces a homogeneous equation for infinitesimal deformations. Expanding off shell retains the linear residual and produces an inhomogeneous equation for a correction.

The Newton interpretation

In finite dimensions, to solve

$$\nabla f(x) = 0 \quad (81)$$

from a nonstationary point $\bar{x}$, one expands

$$\nabla f(\bar{x} + h) = \nabla f(\bar{x}) + H_f(\bar{x})h + O(\|h\|^2). \quad (82)$$

Cancelling the residual to first order gives

$$H_f(\bar{x})h = -\nabla f(\bar{x}). \quad (83)$$

The field-theoretic equation

$$DE_{\bar{g}}[h] = -E[\bar{g}] \quad (84)$$

has exactly the same form. This interpretation also supplies an important limitation. A small correction h can cancel the residual only when the residual is sufficiently small and lies, to the required accuracy, in the image of $DE_{\bar{g}}$. If R is large, singular or contains a component outside that image, the linear Newton step is not a reliable approximation.

The roles should therefore remain distinct:

$R = E[\bar{g}]$	the failure of the background to satisfy the full field equation
$DE_{\bar{g}}$	the linear response of the field-equation residual to a change in the metric
h	the first correction sought for the metric
$\ker DE_{\bar{g}}$	directions which do not change the residual to first order
$\text{im } DE_{\bar{g}}$	residual directions which can be reached by a first-order metric correction

An off-shell residual can arise because the background is a poor approximation, because matter has been omitted, because the proposed field equation is incomplete, or because observations, sources or boundary data have been misrepresented. The residual alone does not distinguish among these possibilities.

6 Gauge directions and the linear deformation problem

The equation

$$DE_{\bar{g}}[h] = 0 \quad (85)$$

selects perturbations which preserve the Einstein equation to first order. Its solutions are not yet all physically distinct. Some arise only because the same geometry can be described in different coordinate systems.

Diffeomorphism covariance of the Einstein map means that

$$E[\varphi^* g] = \varphi^* E[g] \quad (86)$$

for every diffeomorphism φ . Thus transforming the metric transforms the field-equation residual in the same geometrical way. Let φ_ε be a one-parameter family of diffeomorphisms generated by a vector field ξ . Differentiating the covariance relation at $\varepsilon = 0$ gives

$$DE_{\bar{g}}[\mathcal{L}_\xi \bar{g}] = \mathcal{L}_\xi E[\bar{g}]. \quad (87)$$

If the background is on shell, then

$$E[\bar{g}] = 0, \quad (88)$$

and consequently

$$DE_{\bar{g}}[\mathcal{L}_\xi \bar{g}] = 0. \quad (89)$$

The metric perturbation

$$h = \mathcal{L}_\xi \bar{g} \quad (90)$$

therefore solves the linearised equation. It represents an infinitesimal change of description generated by a diffeomorphism, rather than a new spacetime geometry.

Define now the infinitesimal gauge map

$$K_{\bar{g}} : \xi \longmapsto \mathcal{L}_\xi \bar{g}. \quad (91)$$

On an on-shell background, the covariance identity becomes

$$DE_{\bar{g}} \circ K_{\bar{g}} = 0. \quad (92)$$

Hence there is a short linear sequence

$$\Gamma(TM) \xrightarrow{K_{\bar{g}}} \Gamma(S^2 T^* M) \xrightarrow{DE_{\bar{g}}} \Gamma(S^2 T^* M), \quad (93)$$

in which

$$\text{im } K_{\bar{g}} \subseteq \ker DE_{\bar{g}}. \quad (94)$$

The three spaces have different meanings:

Space	Interpretation
$\text{im } K_{\bar{g}}$	infinitesimal changes generated by diffeomorphisms; these are gauge directions
$\ker DE_{\bar{g}}$	all perturbations which satisfy the linearised field equation
$\ker DE_{\bar{g}} / \text{im } K_{\bar{g}}$	infinitesimal solutions after gauge-equivalent perturbations have been identified

Let

$$\mathcal{E} = \{g : E[g] = 0\} \quad (95)$$

be the space of exact solutions. At a sufficiently regular background,

$$T_{\bar{g}}\mathcal{E} \simeq \ker DE_{\bar{g}}. \quad (96)$$

After quotienting by diffeomorphisms, the infinitesimal moduli space is described schematically by

$$\boxed{T_{[\bar{g}]}(\mathcal{E} / \text{Diff}(M)) \simeq \frac{\ker DE_{\bar{g}}}{\text{im } K_{\bar{g}}}.} \quad (97)$$

The brackets $[\bar{g}]$ denote the entire diffeomorphism-equivalence class of the metric. This description is schematic because the action of $\text{Diff}(M)$ need not be free, the differential operators require choices of function spaces and boundary conditions, and backgrounds with Killing fields can be singular points of the quotient.

The kernel of the linearised Einstein operator contains both physical infinitesimal deformations and changes of geometrical description. Gauge reduction separates these by quotienting the kernel by the image of the infinitesimal diffeomorphism map.

Image and cokernel

The same operator also appeared in the off-shell correction equation

$$DE_{\bar{g}}[h] = -R. \quad (98)$$

This equation can be solved only if the residual R lies in the image of $DE_{\bar{g}}$. Thus

$$\text{im } DE_{\bar{g}} \quad (99)$$

contains the residual directions reachable by a first-order change of the metric.

Residuals which cannot be produced in this way are represented by the cokernel,

$$\text{coker } DE_{\bar{g}} = \frac{\text{space of residuals}}{\text{im } DE_{\bar{g}}}. \quad (100)$$

A nonzero class in the cokernel signals a failure of the corresponding linear equation to possess a solution. Under suitable boundary conditions and pairings, the cokernel may be related to the kernel of the formal adjoint:

$$\text{coker } DE_{\bar{g}} \simeq \ker(DE_{\bar{g}})^\dagger. \quad (101)$$

This is the setting in which solvability conditions and obstruction projections are usually formulated.

7 From infinitesimal deformations to exact solutions

Solving

$$DE_{\bar{g}}[h] = 0 \quad (102)$$

shows that h preserves the field equation to first order. It does not yet show that h is tangent to an actual family of exact solutions. The next question is therefore:

Can the infinitesimal deformation h be continued to second and higher orders?

Suppose there exists a family

$$g(\varepsilon) = \bar{g} + \varepsilon h + \frac{\varepsilon^2}{2} k + O(\varepsilon^3) \quad (103)$$

satisfying

$$E[g(\varepsilon)] = 0. \quad (104)$$

Taylor expansion of the Einstein map gives

$$\begin{aligned} E[g(\varepsilon)] &= E[\bar{g}] + \varepsilon DE_{\bar{g}}[h] \\ &\quad + \frac{\varepsilon^2}{2} \left(DE_{\bar{g}}[k] + D^2 E_{\bar{g}}[h, h] \right) + O(\varepsilon^3). \end{aligned} \quad (105)$$

The background and first-order equations have already supplied

$$E[\bar{g}] = 0, \quad DE_{\bar{g}}[h] = 0. \quad (106)$$

The first potentially nonzero term is therefore

$$E[g(\varepsilon)] = \frac{\varepsilon^2}{2} \left(DE_{\bar{g}}[k] + D^2 E_{\bar{g}}[h, h] \right) + O(\varepsilon^3). \quad (107)$$

The factor $\frac{1}{2}$ multiplies the entire second-order coefficient. Requiring $E[g(\varepsilon)] = 0$ through second order therefore gives

$$DE_{\bar{g}}[k] + D^2 E_{\bar{g}}[h, h] = 0. \quad (108)$$

Here k is the second-order correction. The known first-order deformation h produces the quadratic source

$$D^2 E_{\bar{g}}[h, h], \quad (109)$$

and k must cancel it through the linear operator $DE_{\bar{g}}$:

$$DE_{\bar{g}}[k] = -D^2 E_{\bar{g}}[h, h]. \quad (110)$$

This equation can be solved only if

$$D^2 E_{\bar{g}}[h, h] \in \text{im } DE_{\bar{g}}. \quad (111)$$

Equivalently, its class in the cokernel must vanish:

$$\boxed{[D^2 E_{\bar{g}}[h, h]] = 0 \quad \text{in} \quad \text{coker } DE_{\bar{g}}.} \quad (112)$$

Thus $D^2 E_{\bar{g}}[h, h]$ is not by itself an obstruction. Its projection onto the cokernel is the obstruction. A nonzero projection means that h solves the linearised equation but cannot be continued to second order. Under a suitable pairing, let ψ lie in

$$\ker(DE_{\bar{g}})^\dagger. \quad (113)$$

Under the boundary or fall-off conditions stated above, let

$$\psi \in \ker(DE_{\bar{g}})^\dagger. \quad (114)$$

Using the L^2 -type pairing on symmetric rank-two tensors defined earlier, a corresponding solvability condition is

$$\langle \psi, D^2 E_{\bar{g}}[h, h] \rangle_{\bar{g}} = 0. \quad (115)$$

This condition requires the quadratic source to be orthogonal to every element of the adjoint kernel. If it fails for some ψ , then $D^2 E_{\bar{g}}[h, h]$ has a nonzero component in $\text{coker } DE_{\bar{g}}$ which no choice of the second-order correction k can remove.

The kernel determines candidate infinitesimal deformations. The cokernel tests whether the nonlinear source generated by those deformations can be cancelled at the next order.

The derivative hierarchy

Since the Euler–Lagrange map is the first derivative of the action,

$$E = DS, \quad (116)$$

the successive derivatives are related by

$$DE = D^2 S, \quad D^2 E = D^3 S, \quad D^3 E = D^4 S, \quad \dots \quad (117)$$

Their roles in the deformation problem are

$$\begin{aligned} E[\bar{g}] &= 0 && \text{selects the exact background,} \\ DE_{\bar{g}}[h] &= 0 && \text{selects candidate infinitesimal deformations,} \\ [D^2 E_{\bar{g}}[h, h]] &= 0 && \text{tests their continuation to second order,} \\ D^3 E_{\bar{g}}, D^4 E_{\bar{g}}, \dots &&& \text{provide higher compatibility data.} \end{aligned} \quad (118)$$

These equations do not require every derivative of the action to vanish. Rather, each order supplies an equation for the next coefficient in the prospective family $g(\varepsilon)$. At second order, for example, $D^3 S$ provides a source which the Hessian $D^2 S$ must absorb through the correction k .

An infinitesimal solution which extends to an exact family is called *integrable*. One which satisfies the linearised equation but fails a higher-order solvability condition is obstructed. This distinction is central to the linearisation-stability problem for Einstein’s equations [3]. Concrete second-order Einstein deformations exhibit precisely this passage from kernel elements to quadratic obstruction classes [4].

Equivariance, deformation and formal integrability

The deformation problem is equivariant under the diffeomorphism group. The group action determines which perturbations are gauge-equivalent, while the Einstein map determines the invariant solution set. Its first derivative supplies the linear deformation operator, and its higher derivatives supply the nonlinear terms entering the obstruction equations.

This is the precise contribution of the Erlangen perspective: geometrical structures are considered together with the transformations under which they are equivalent. The group action organizes the quotient

$$\mathcal{E}/\text{Diff}(M), \quad (119)$$

but the group principle alone does not supply the full deformation or obstruction theory. Those require the differential structure of the Einstein map and its successive derivatives.

A complementary jet-bundle description studies the differential equation together with its prolongations. It asks whether solutions of the equations at one jet order extend consistently to higher jet orders, and whether new compatibility conditions appear. This is formal integrability in the jet sense [1]. The two hierarchies address related continuation questions from different directions:

Variational deformation hierarchy	Expands a family in a parameter ε and asks whether an infinitesimal deformation extends to a nearby exact solution
Jet-prolongation hierarchy	Raises the differential order and asks whether lower-order equation data extend consistently to higher jets

They are complementary rather than identical. Deformation theory concerns continuation in the space of fields or solutions; jet theory concerns compatibility across differential order. In metric-affine gravity, where the metric and connection may be varied independently, the jet-bundle formulation can make the differential compatibility structure especially explicit.

8 Effective sources as representation

Suppose a modified linearised equation is proposed:

$$(\mathcal{H}_E + \Delta\mathcal{H})h = \kappa \delta T_{\text{vis}}. \quad (120)$$

It may always be rewritten as

$$\mathcal{H}_E h = \kappa (\delta T_{\text{vis}} + \delta T_{\text{eff}}[h]), \quad (121)$$

where

$$\delta T_{\text{eff}}[h] = -\frac{1}{\kappa} \Delta\mathcal{H}h. \quad (122)$$

This rewriting is mathematically valid. It does not decide whether the new term represents matter, modified gravitational dynamics, an approximation, or a field parametrisation. Calling it an effective dark-matter tensor does not supply an independent equation of state, particle dynamics, clustering law or observational signature.

Likewise, an off-shell residual may be written

$$T_{\mu\nu}^{\text{eff}} = -\kappa^{-1} E_{\mu\nu}[\bar{g}]. \quad (123)$$

This answers a limited inverse question: which stress tensor would make the chosen metric satisfy the Einstein equation? It does not identify the physical origin of that stress tensor.

An operator term moved to the right-hand side has acquired the grammatical position of a source. It has not thereby acquired the ontology or independent dynamics of matter.

9 A failed candidate: equations of motion squared

A tempting way to preserve every Einstein background while changing the quadratic response is

$$\Delta S[g] = \frac{\alpha}{2} \langle E[g], \mathcal{A}_g E[g] \rangle. \quad (124)$$

At a solution $E[\bar{g}] = 0$,

$$\Delta S[\bar{g}] = 0, \quad \delta \Delta S_{\bar{g}} = 0, \quad (125)$$

while

$$\delta^2 \Delta S_{\bar{g}}[h, k] = \alpha \langle DE_{\bar{g}}[h], \mathcal{A}_{\bar{g}} DE_{\bar{g}}[k] \rangle. \quad (126)$$

Formally,

$$\Delta \mathcal{H} = \alpha (DE_{\bar{g}})^\dagger \mathcal{A}_{\bar{g}} DE_{\bar{g}}. \quad (127)$$

The construction appears to keep the stationary locus while altering its sourced response. Two difficulties undermine that interpretation.

Perturbative field-redefinition redundancy

Consider the field redefinition

$$g \longmapsto g + \alpha \mathcal{A}_g E[g]. \quad (128)$$

To first order in α ,

$$S[g + \alpha \mathcal{A}_g E[g]] = S[g] + \alpha \langle E[g], \mathcal{A}_g E[g] \rangle + O(\alpha^2), \quad (129)$$

up to conventions, boundary terms and the transformation of matter couplings. An equations-of-motion-squared operator is therefore generally redundant at this order in an effective-field-theory expansion. It may change the appearance of the quadratic operator without changing perturbative observables. This is the standard redundancy of operators proportional to lower-order equations of motion, subject to the stated qualifications concerning sources, boundaries and observables [5].

Higher derivatives and additional modes

The Einstein equation already contains second derivatives of the metric. A local term quadratic in $E_{\mu\nu}$ generically produces fourth-order field equations when treated as part of an exact theory. The additional derivatives can introduce new poles and propagating degrees of freedom. Generic curvature-squared gravity contains a problematic massive spin-2 mode; the precise spectrum depends on the invariant combination, degeneracies, constraints, dimension, boundary conditions and whether the theory is treated only as a low-energy expansion. The extra massive spin-2 pole and its wrong-sign residue in generic higher-derivative gravity are classic results [6, 7]; they are a warning about a nonperturbative completion, not a claim that every effective higher-curvature term is by itself inconsistent. Thus the same candidate faces two different verdicts in two regimes:

Perturbative EFT treatment	frequently removable at leading order by a field redefinition, so it supplies no new leading physical effect
Exact higher-derivative treatment	may introduce additional modes, including ghostlike instabilities in generic cases

The example remains useful as a failed construction. It demonstrates that preserving a stationary locus while modifying its Hessian is too weak a criterion for new physics. One must also establish non-redundancy, a consistent nonlinear completion, an acceptable spectrum and distinct observable predictions.

10 Conclusion

The chain is thus

$$S \longrightarrow E = DS \longrightarrow DE = D^2S \longrightarrow D^2E = D^3S \longrightarrow \dots . \quad (130)$$

The first variation selects a stationary metric. The Jacobian of the Einstein map, equivalently the Hessian of the action, selects infinitesimal deformations. Higher derivatives test their nonlinear continuation. Diffeomorphism equivariance organizes the gauge directions throughout this hierarchy.

An off-shell residual is valuable because it measures failure of stationarity and sources a first Newton correction. It does not identify the physical cause of that failure. Similarly, an effective source obtained by moving an operator term across the field equation is a representation of modified response. Its physical interpretation requires independent dynamics and observables. In sum we have:

The Hessian of the action is the tangent map of its stationary equation. Its kernel describes infinitesimal stationary structure; its image controls correctable residuals; its co-kernel carries possible obstructions.

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